

Initiating Search

November 10, 2025, 4:40 PM

• Search:

CAS Lexicon Search:

Trifluoromethylation - Preferred Concept

Search Tasks

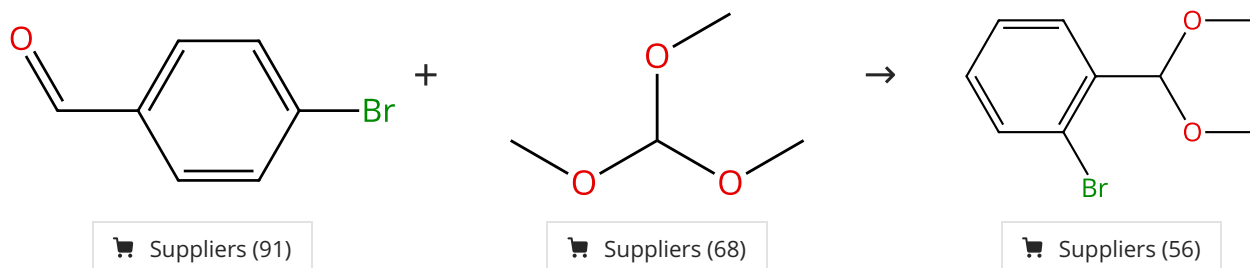
Task	Result Type	View
Returned Reference Results from CAS Lexicon (2,963)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (424)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Non-Participating Functional Groups:	Halide, Alkyl halide, Alkene, Cyclic alkene, Amine, Diene, Amide, Phenyl halide, Primary amine, Secondary amine, Imine	
Document Type:	Journal	

Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-085-CAS-16907049

Steps: 1 Yield: 100%

Synthesis of Trifluoromethylated Naphthoquinones via Copper-Catalyzed Cascade Trifluoromethylation/Cyclization of 2-(3-Arylpropiolyl)benzaldehydes

1.1 Catalysts: *p*-Toluenesulfonic acid
Solvents: Methanol; 8 h, 70 °C

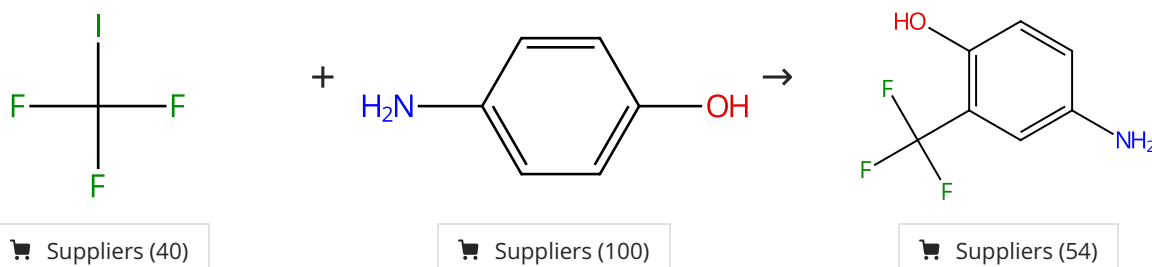
Experimental Protocols

By: Zhang, Yan; et al

Organic Letters (2017), 19(6), 1302-1305.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



31-085-CAS-16579226

Steps: 1 Yield: 100%

Continuous Flow Homolytic Aromatic Substitution with Electrophilic Radicals: A Fast and Scalable Protocol for Trifluoromethylation

1.1 Reagents: Sulfuric acid
Catalysts: Ferrous sulfate
Solvents: Dimethyl sulfoxide, Acetonitrile, Water

1.2 Reagents: Hydrogen peroxide
Solvents: Water; 22 s, rt

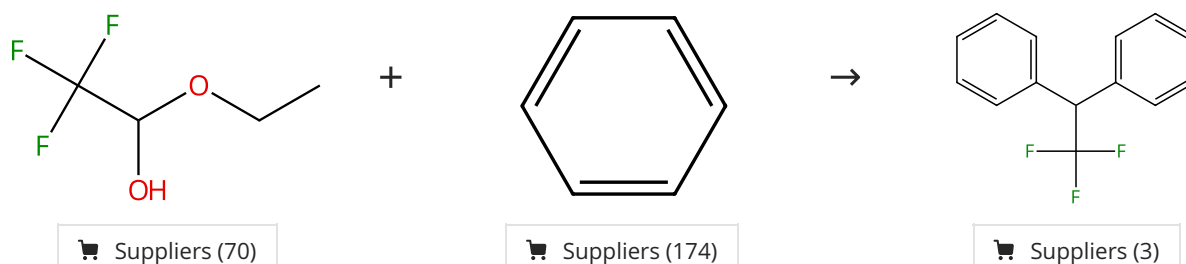
1.3 Reagents: Sodium thiosulfate
Solvents: Water; rt

By: Monteiro, Julia L.; et al

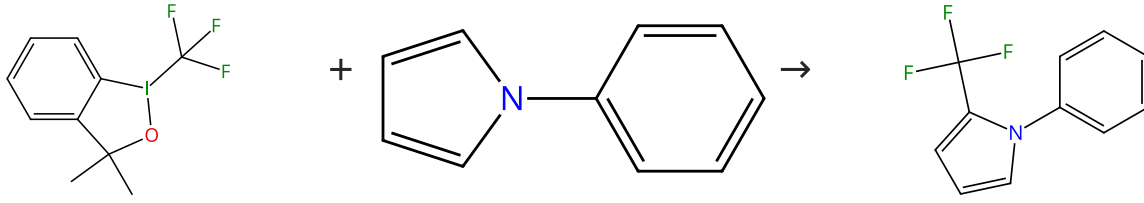
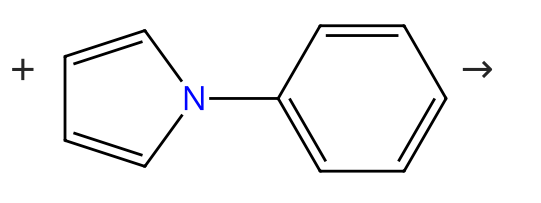
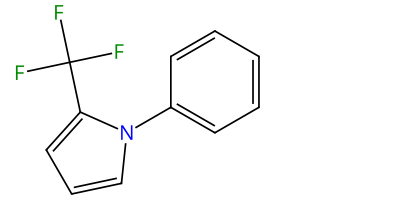
Chemistry - A European Journal (2017), 23(1), 176-186.

Scheme 3 (1 Reaction)

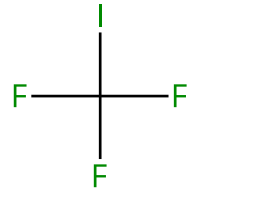
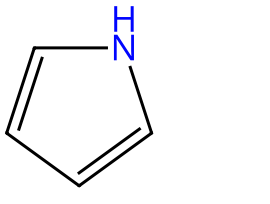
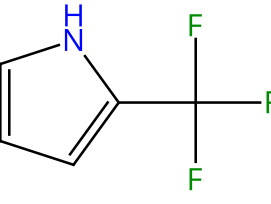
Steps: 1 Yield: 100%



31-085-CAS-3735017	Steps: 1 Yield: 100%	Fluoral hemiacetal, a new reagent for aromatic trifluoro alkylation By: Guy, A.; et al Journal of Fluorine Chemistry (1986), 32(4), 361-6.
1.1 Catalysts: Sulfuric acid Solvents: Dichloromethane		

Scheme 4 (1 Reaction) Steps: 1 Yield: 99%		
		
 Suppliers (67)	 Suppliers (71)	 Supplier (1)

31-085-CAS-2078284	Steps: 1 Yield: 99%	Electrophilic trifluoromethylation of arenes and N-hetero arenes using hypervalent iodine reagents By: Wiehn, Matthias S.; et al Journal of Fluorine Chemistry (2010), 131(9), 951-957.
1.1 Reagents: (7-4)-Bis[1,1,1-trifluoro-N-[(trifluoromethyl)sulfonyl-κO]methanesulfonamidato-κO]zinc Solvents: Chloroform; 24 h, 20 °C		

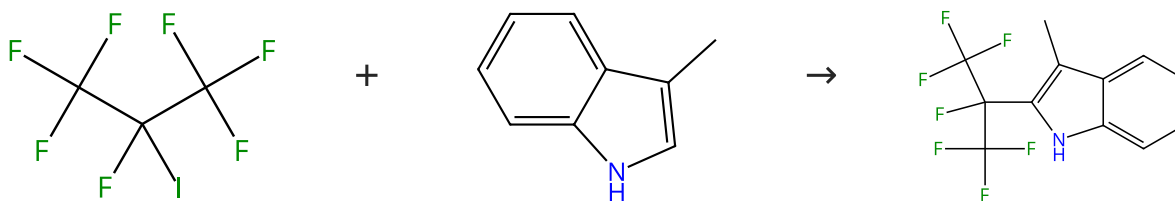
Scheme 5 (2 Reactions) Steps: 1 Yield: 96-99%		
		
 Suppliers (40)	 Suppliers (76)	 Suppliers (34)

31-085-CAS-18921199	Steps: 1 Yield: 99%	Strongly Reducing, Visible-Light Organic Photoredox Catalysts as Sustainable Alternatives to Precious Metals By: Du, Ya; et al Chemistry - A European Journal (2017), 23(46), 10962-10968.
1.1 Reagents: Potassium formate Catalysts: 5,10-Dihydro-5,10-di-2-naphthalenylphenazine Solvents: Dimethylacetamide; 6 h, 40 °C		
1.2 Solvents: Water		
Experimental Protocols		

31-085-CAS-626801	Steps: 1 Yield: 96%	Trifluoromethylation of various aromatic compounds by C F ₃ I in the presence of Fe(II) compound, H ₂ O ₂ and dimethyl sulfoxide By: Kino, Tatsuhito; et al Journal of Fluorine Chemistry (2010), 131(1), 98-105.
1.1 Reagents: Dimethyl sulfoxide, Sulfuric acid, Ferrous sulfate, Hydrogen peroxide Solvents: Dimethyl sulfoxide, Water; rt → 50 °C; 20 min, 40 - 50 °C; 50 °C → rt		
1.2 Reagents: Sodium carbonate Solvents: Water; neutralized, rt		

Scheme 6 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (58)

Suppliers (105)

Suppliers (3)

31-085-CAS-10867666

Steps: 1 Yield: 99%

Rapid trifluoromethylation and perfluoroalkylation of five-membered heterocycles by photoredox catalysis in continuous flow

By: Straathof, Natan J. W.; et al

ChemSusChem (2014), 7(6), 1612-1617.

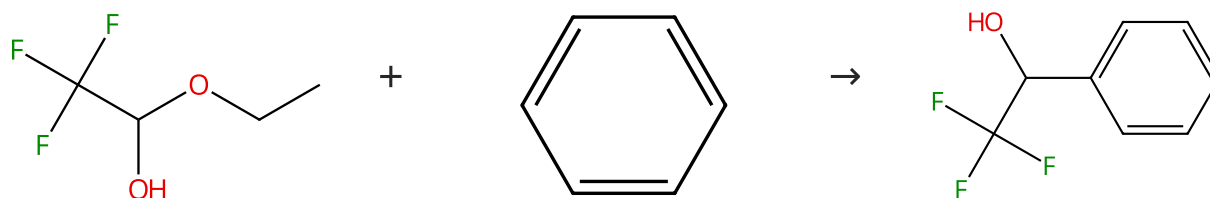
1.1 **Catalysts:** 1,3-Propanediamine, Tris(2,2'-bipyridyl)ruthenium(I) chloride

Solvents: Acetonitrile; 10 min

1.2 -

Scheme 7 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (70)

Suppliers (174)

Suppliers (80)

31-085-CAS-14739467

Steps: 1 Yield: 98%

Fluoral hemiacetal, a new reagent for aromatic trifluoro alkylation

By: Guy, A.; et al

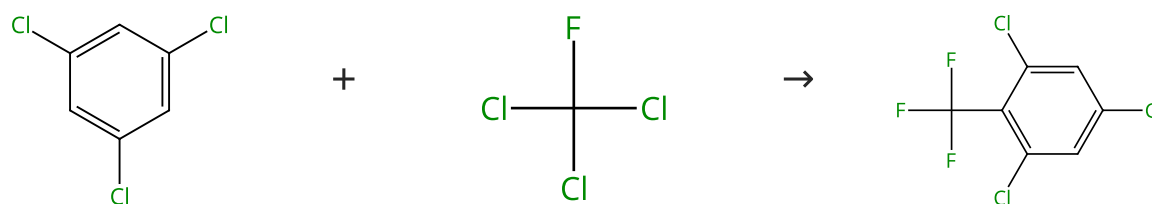
Journal of Fluorine Chemistry (1986), 32(4), 361-6.

1.1 **Catalysts:** Aluminum chloride

Solvents: Dichloromethane

Scheme 8 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (56)

Suppliers (27)

Suppliers (18)

31-046-CAS-11484343

Steps: 1 Yield: 98%

New synthesis of polychloro(trifluoromethyl)benzenes and highly strained polychloro(trichloromethyl)benzenes

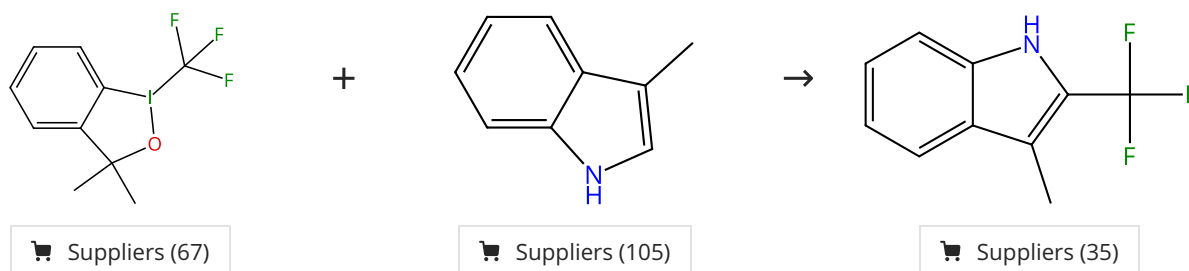
By: Riera, Juan; et al

Tetrahedron Letters (1989), 30(29), 3825-8.

1.1 **Reagents:** Aluminum chloride

Scheme 9 (1 Reaction)

Steps: 1 Yield: 98%



31-085-CAS-8473752

Steps: 1 Yield: 98%

Electrophilic trifluoromethylation of arenes and N-hetero arenes using hypervalent iodine reagents

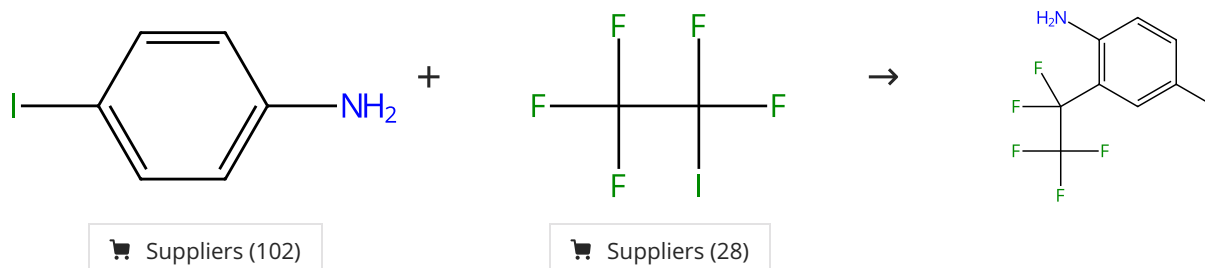
1.1 **Reagents:** (7-4)-Bis[1,1,1-trifluoro-N-[(trifluoromethyl)sulfonyl-κO]methanesulfonamidato-κO]zinc
Solvents: Acetonitrile; 24 h, 80 °C

By: Wiehn, Matthias S.; et al

Journal of Fluorine Chemistry (2010), 131(9), 951-957.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 98%



31-085-CAS-18338690

Steps: 1 Yield: 98%

Visible-light-mediated direct perfluoroalkylation and trifluoromethylation of free anilines

1.1 **Reagents:** Potassium carbonate
Catalysts: *fac*-Tris(2-(2-pyridinyl)phenyl)iridium
Solvents: 1,2-Dichloroethane; 24 h, rt

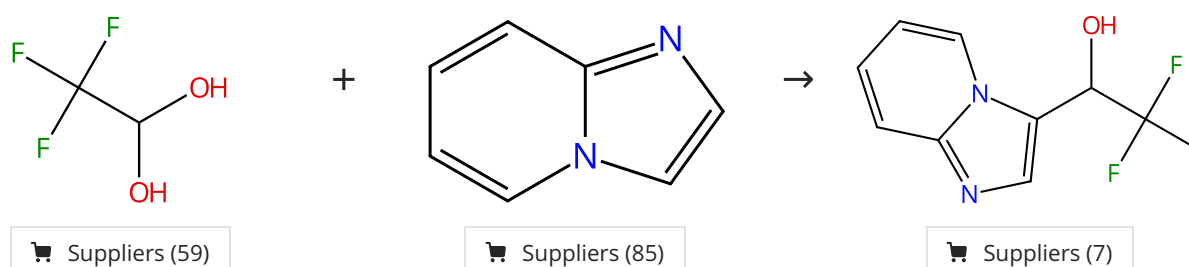
By: He, Chun-Yang; et al

Tetrahedron Letters (2017), 58(41), 3939-3941.

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-39521361

Steps: 1 Yield: 97%

Metal-Free Synthesis of Trifluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines via Dehydrative Coupling of Imidazo[1,2-a]pyridines with Trifluoroacetaldehyde
1.1 **Solvents:** 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt1.2 **Solvents:** Water; rt

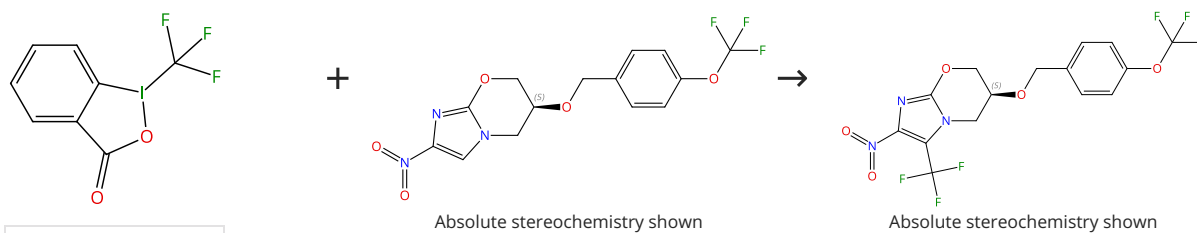
By: Liu, Zhaowen; et al

Synthesis (2024), 56(11), 1756-1764.

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (65)

Suppliers (78)

31-614-CAS-31393791

Steps: 1 Yield: 97%

Transition-metal-free, direct C-H radical trifluoromethylation of nitroimidazoles with Togni's reagent

1.1 **Reagents:** Potassium acetate
Solvents: Dimethyl sulfoxide; 24 h, rt

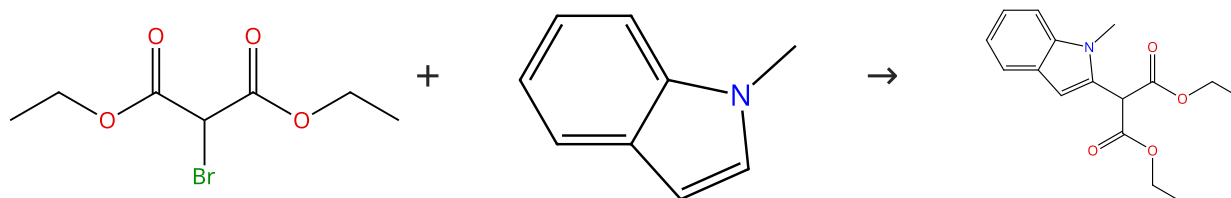
By: Wang, Hezhen; et al

Experimental Protocols

Tetrahedron Letters (2022), 92, 153659.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (74)

Suppliers (102)

Supplier (1)

31-085-CAS-19937752

Steps: 1 Yield: 97%

De novo Design of Organic Photocatalysts: Bithiophene Derivatives for the Visible-light Induced C-H Functionalization of Heteroarenes

1.1 **Reagents:** Triphenylamine
Catalysts: 5,5'-Bis[2-[2-(trifluoromethyl)phenyl]ethynyl]-2,2'-bithiophene
Solvents: Dimethylformamide; 10 min, rt

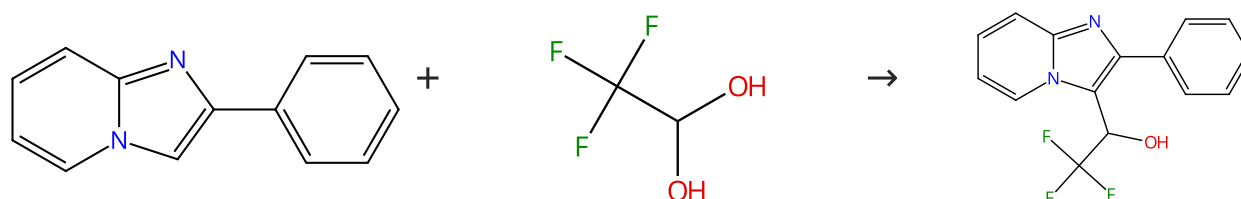
By: Bottecchia, Cecilia; et al

Experimental Protocols

Advanced Synthesis & Catalysis (2019), 361(5), 945-950.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (80)

Suppliers (59)

31-614-CAS-39521364

Steps: 1 Yield: 96%

Metal-Free Synthesis of Trifluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines via Dehydrative Coupling of Imidazo[1,2-a]pyridines with Trifluoroacetaldehyde
1.1 **Solvents:** 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt1.2 **Solvents:** Water; rt

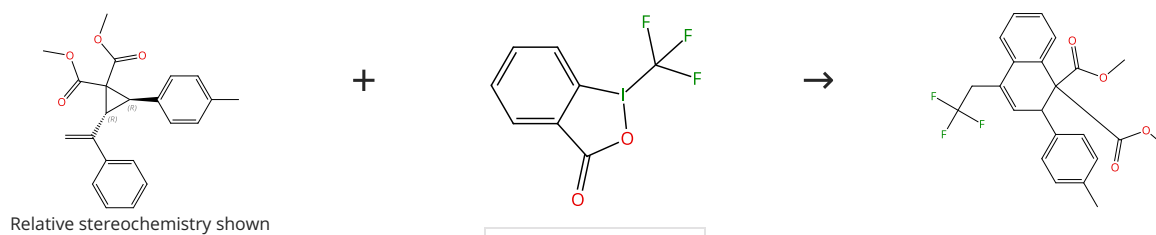
By: Liu, Zhaowen; et al

Experimental Protocols

Synthesis (2024), 56(11), 1756-1764.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (65)

31-085-CAS-17104265

Steps: 1 Yield: 96%

Iron-catalyzed trifluoromethylation of vinyl cyclopropanes: facile synthesis of CF₃-containing dihydronaphthalene derivatives

By: Li, Zong-Rui; et al

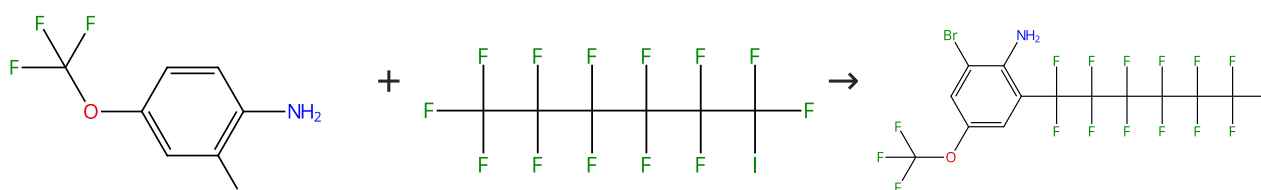
Organic Chemistry Frontiers (2016), 3(8), 934-938.

- 1.1 **Catalysts:** Ferrous chloride
Solvents: Dichloromethane; 10 - 30 min, 40 °C
- 1.2 **Reagents:** Potassium carbonate
Solvents: Water

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (77)

Suppliers (74)

31-085-CAS-18338687

Steps: 1 Yield: 96%

Visible-light-mediated direct perfluoroalkylation and trifluoromethylation of free anilines

By: He, Chun-Yang; et al

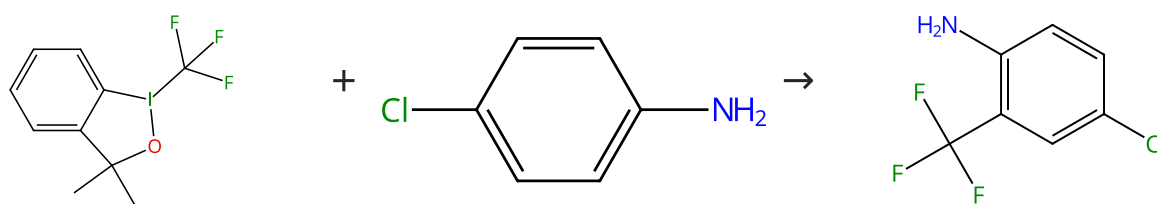
Tetrahedron Letters (2017), 58(41), 3939-3941.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: *fac*-Tris(2-(2-pyridinyl)phenyl)iridium
Solvents: 1,2-Dichloroethane; 24 h, rt

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (67)

Suppliers (98)

Suppliers (89)

31-085-CAS-22065856

Steps: 1 Yield: 96%

Transition metal-free direct C-H trifluoromethylation of (hetero)arenes with Togni's reagent

By: Chen, Xiaoyu; et al

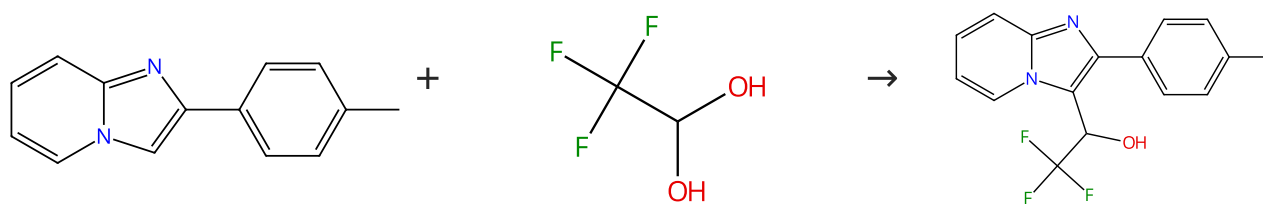
Tetrahedron Letters (2020), 61(9), 151538.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Acetonitrile; 6 h, 75 °C

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (53)

Suppliers (59)

31-614-CAS-39521358

Steps: 1 Yield: 95%

Metal-Free Synthesis of Trifluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines via Dehydrative Coupling of Imidazo[1,2-a]pyridines with Trifluoroacetaldehyde

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

1.2 Solvents: Water; rt

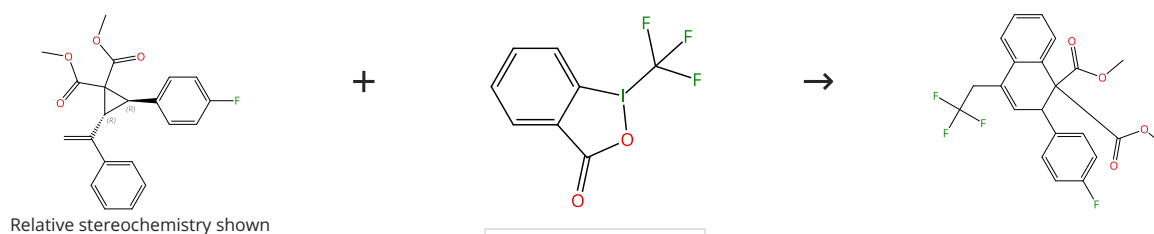
Experimental Protocols

By: Liu, Zhaowen; et al

Synthesis (2024), 56(11), 1756-1764.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (65)

31-085-CAS-17104260

Steps: 1 Yield: 95%

Iron-catalyzed trifluoromethylation of vinyl cyclopropanes: facile synthesis of CF₃-containing dihydronaphthalene derivatives

1.1 Catalysts: Ferrous chloride

Solvents: Dichloromethane; 10 - 30 min, 40 °C

1.2 Reagents: Potassium carbonate

Solvents: Water

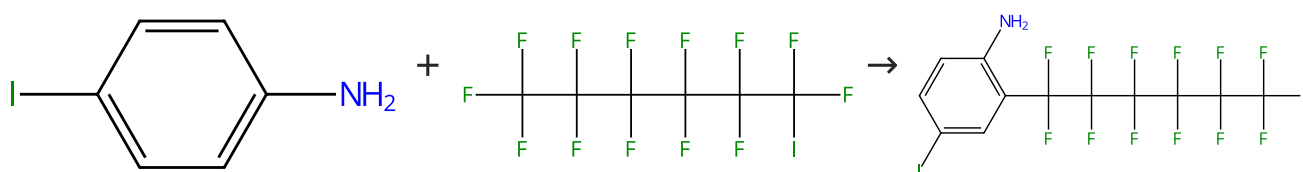
Experimental Protocols

By: Li, Zong-Rui; et al

Organic Chemistry Frontiers (2016), 3(8), 934-938.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (102)

Suppliers (74)

31-085-CAS-18336799

Steps: 1 Yield: 95%

Visible-light-mediated direct perfluoroalkylation and trifluoromethylation of free anilines

1.1 Reagents: Potassium carbonate

Catalysts: *fac*-Tris(2-(2-pyridinyl)phenyl)iridium

Solvents: 1,2-Dichloroethane; 24 h, rt

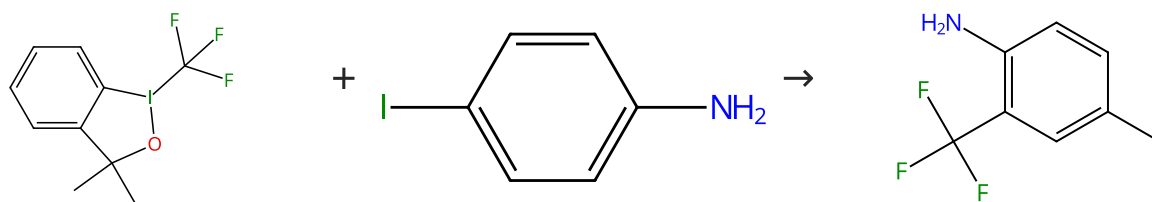
Experimental Protocols

By: He, Chun-Yang; et al

Tetrahedron Letters (2017), 58(41), 3939-3941.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (67)

Suppliers (102)

Suppliers (55)

31-085-CAS-22065854

Steps: 1 Yield: 95%

Transition metal-free direct C-H trifluoromethylation of (hetero)arenes with Togni's reagent

By: Chen, Xiaoyu; et al

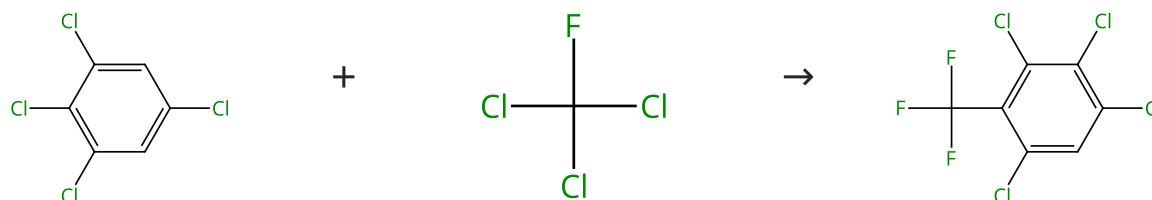
Tetrahedron Letters (2020), 61(9), 151538.

 1.1 Reagents: Potassium carbonate
 Solvents: Acetonitrile; 6 h, 75 °C

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (60)

Suppliers (27)

Suppliers (5)

31-046-CAS-8960011

Steps: 1 Yield: 95%

New synthesis of polychloro(trifluoromethyl)benzenes and highly strained polychloro(trichloromethyl)benzenes

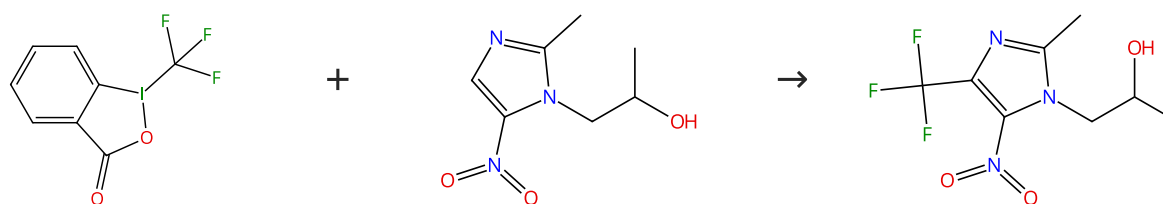
By: Riera, Juan; et al

Tetrahedron Letters (1989), 30(29), 3825-8.

1.1 Reagents: Aluminum chloride

Scheme 23 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (65)

Suppliers (88)

31-614-CAS-31393763

Steps: 1 Yield: 95%

Transition-metal-free, direct C-H radical trifluoromethylation of nitroimidazoles with Togni's reagent

By: Wang, Hezhen; et al

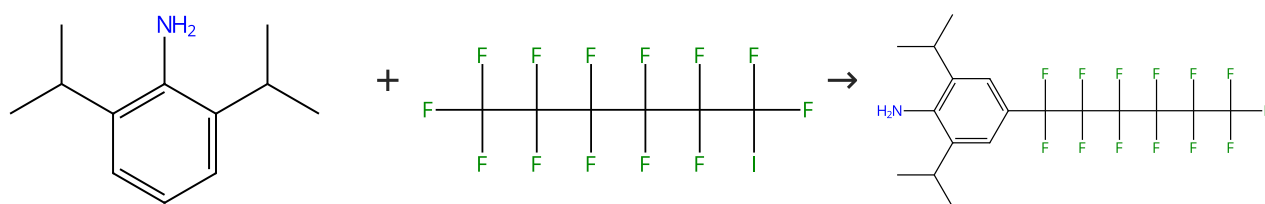
Tetrahedron Letters (2022), 92, 153659.

 1.1 Reagents: Potassium acetate
 Solvents: Dimethyl sulfoxide; 24 h, rt

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (90)

Suppliers (74)

31-085-CAS-18338689

Steps: 1 Yield: 95%

Visible-light-mediated direct perfluoroalkylation and trifluoromethylation of free anilines

1.1 **Reagents:** Potassium carbonate
Catalysts: *fac*-Tris(2-(2-pyridinyl)phenyl)iridium
Solvents: 1,2-Dichloroethane; 24 h, rt

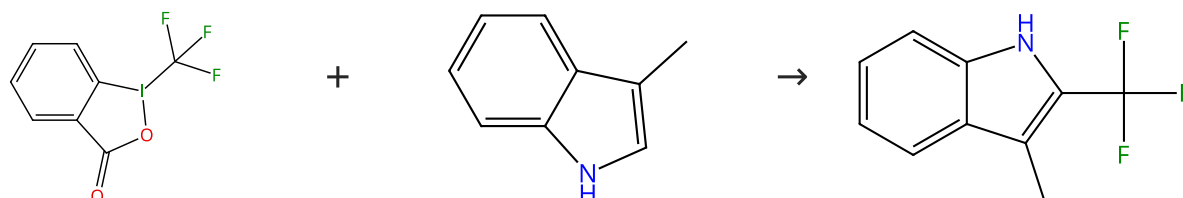
By: He, Chun-Yang; et al

Tetrahedron Letters (2017), 58(41), 3939-3941.

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (65)

Suppliers (105)

Suppliers (35)

31-085-CAS-6739035

Steps: 1 Yield: 95%

Direct C2-trifluoromethylation of indole derivatives catalyzed by copper acetate

1.1 **Catalysts:** Cuprous acetate
Solvents: Methanol; 6 h, rt

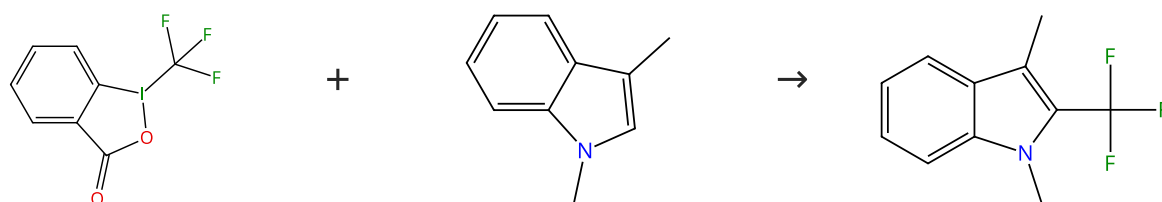
By: Shimizu, Ryo; et al

Tetrahedron Letters (2010), 51(45), 5947-5949.

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (65)

Suppliers (62)

Suppliers (4)

31-085-CAS-465817

Steps: 1 Yield: 95%

Direct C2-trifluoromethylation of indole derivatives catalyzed by copper acetate

1.1 **Catalysts:** Cuprous acetate
Solvents: Methanol; 6 h, rt

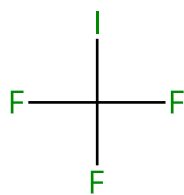
By: Shimizu, Ryo; et al

Tetrahedron Letters (2010), 51(45), 5947-5949.

Experimental Protocols

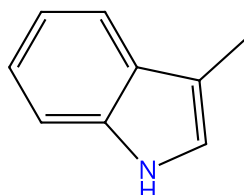
Scheme 27 (1 Reaction)

Steps: 1 Yield: 95%



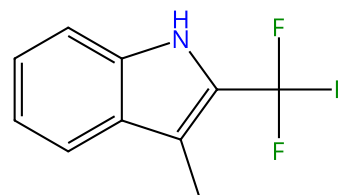
Suppliers (40)

+



Suppliers (105)

→



Suppliers (35)

31-085-CAS-10220405

Steps: 1 Yield: 95%

Rapid trifluoromethylation and perfluoroalkylation of five-membered heterocycles by photoredox catalysis in continuous flow

By: Straathof, Natan J. W.; et al

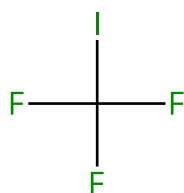
ChemSusChem (2014), 7(6), 1612-1617.

 1.1 **Catalysts:** 1,3-Propanediamine, Tris(2,2'-bipyridyl)ruthenium(II) chloride
Solvents: Acetonitrile; 8 min

1.2 -

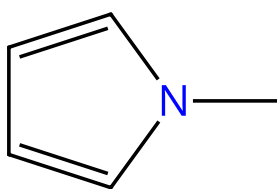
Scheme 28 (4 Reactions)

Steps: 1 Yield: 95%



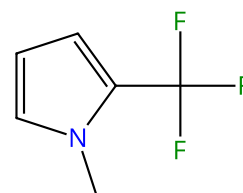
Suppliers (40)

+



Suppliers (75)

→



Suppliers (4)

31-085-CAS-20636862

Steps: 1 Yield: 95%

Visible-Light-Induced Trifluoromethylation of Unactivated Alkenes with Tri(9-anthryl)borane as an Organophotocatalyst

By: Moon, Jisu; et al

Journal of Organic Chemistry (2019), 84(20), 12925-12932.

 1.1 **Reagents:** *N,N,N,N*-Tetramethylethylenediamine
Catalysts: Tri-9-anthracenylborane**Solvents:** Acetonitrile; 6 h, rt

31-085-CAS-12872321

Steps: 1 Yield: 95%

Mechanisms and applications of cyclometalated Pt(II) complexes in photoredox catalytic trifluoromethylation

By: Choi, Won Joon; et al

Chemical Science (2015), 6(2), 1454-1464.

 1.1 **Reagents:** 1,3-Propanediamine
Catalysts: (*SP*-4-3)-(2,4-Pentanedionato-κ^O²,κ^O⁴)[2-(2-pyridinyl-κ^N)phenyl-κ^C]platinum
Solvents: Acetonitrile; 10 - 24 h, rt

31-085-CAS-9186454

Steps: 1 Yield: 95%

Rapid trifluoromethylation and perfluoroalkylation of five-membered heterocycles by photoredox catalysis in continuous flow

By: Straathof, Natan J. W.; et al

ChemSusChem (2014), 7(6), 1612-1617.

 1.1 **Catalysts:** 1,3-Propanediamine, Tris(2,2'-bipyridyl)ruthenium(II) chloride
Solvents: Acetonitrile; 8 min

1.2 -

31-085-CAS-13906653

Steps: 1 Yield: 95%

Trifluoromethylation of heterocycles via visible light photoredox catalysis

By: Iqbal, Naeem; et al

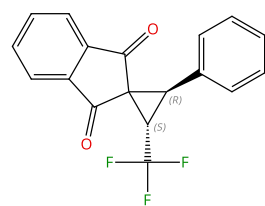
Tetrahedron Letters (2012), 53(15), 2005-2008.

 1.1 **Reagents:** *N,N,N,N*-Tetramethylethylenediamine
Catalysts: Tris(2,2'-bipyridyl)ruthenium(II) chloride
Solvents: Acetonitrile; 10 - 24 h, rt

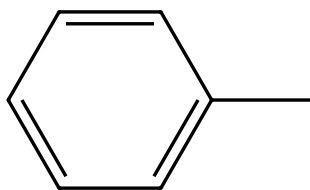
Experimental Protocols

Scheme 29 (1 Reaction)

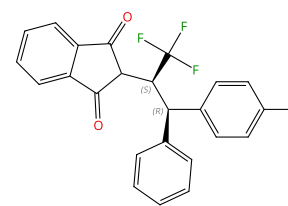
Steps: 1 Yield: 94%



+



→



Suppliers (280)

31-085-CAS-23706633

Steps: 1 Yield: 94%

Additive-Free Synthesis of Trifluoromethylated Spiro Cyclopropanes and Their Transformation into Trifluoromethylated Building Blocks

By: Kumar, Anuj; et al

Asian Journal of Organic Chemistry (2021), 10(6), 1536-1541.

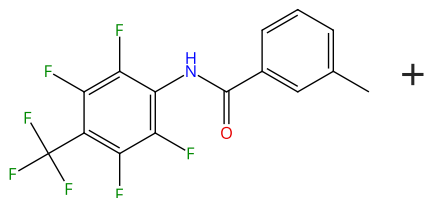
1.1 **Catalysts:** Aluminum chloride
Solvents: 1,2-Dichloroethane; 3 h, 25 °C

1.2 **Reagents:** Water

Experimental Protocols

Scheme 30 (1 Reaction)

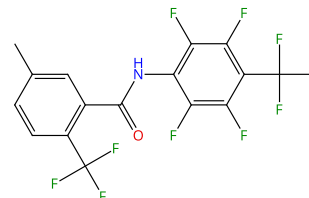
Steps: 1 Yield: 94%



+

Multi-component
 structure image
 available in CAS
 SciFinder

→



Suppliers (70)

31-085-CAS-10143830

Steps: 1 Yield: 94%

Pd(II)-Catalyzed Ortho Trifluoromethylation of Arenes and Insights into the Coordination Mode of Acidic Amide Directing Groups

By: Zhang, Xing-Guo; et al

Journal of the American Chemical Society (2012), 134(29), 11948-11951.

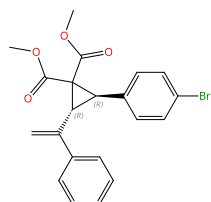
1.1 **Reagents:** Trifluoroacetic acid, *N*-Methylformamide, Cupric acetate

Catalysts: Palladium diacetate
Solvents: 1,2-Dichloroethane; 24 h, 130 °C

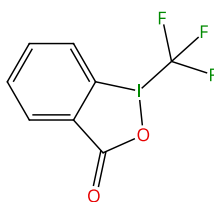
Experimental Protocols

Scheme 31 (1 Reaction)

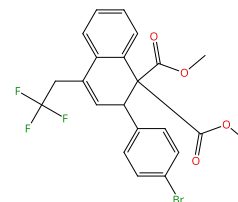
Steps: 1 Yield: 94%



+



→



Suppliers (65)

31-085-CAS-17104262

Steps: 1 Yield: 94%

Iron-catalyzed trifluoromethylation of vinyl cyclopropanes: facile synthesis of CF₃-containing dihydronaphthalene derivatives

By: Li, Zong-Rui; et al

Organic Chemistry Frontiers (2016), 3(8), 934-938.

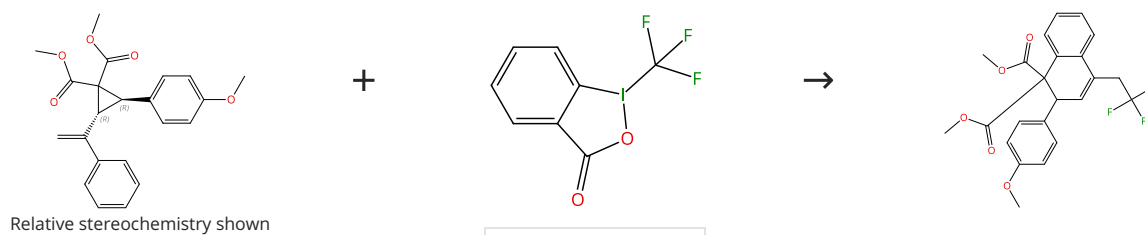
1.1 **Catalysts:** Ferrous chloride
Solvents: Dichloromethane; 10 - 30 min, 40 °C

1.2 **Reagents:** Potassium carbonate
Solvents: Water

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (65)

31-085-CAS-17104264

Steps: 1 Yield: 94%

Iron-catalyzed trifluoromethylation of vinyl cyclopropanes: facile synthesis of CF₃-containing dihydronaphthalene derivatives

By: Li, Zong-Rui; et al

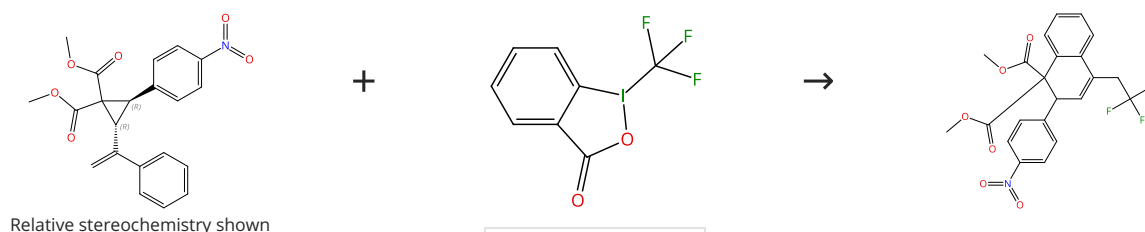
Organic Chemistry Frontiers (2016), 3(8), 934-938.

- 1.1 **Catalysts:** Ferrous chloride
Solvents: Dichloromethane; 10 - 30 min, 40 °C
- 1.2 **Reagents:** Potassium carbonate
Solvents: Water

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (65)

31-085-CAS-17104263

Steps: 1 Yield: 94%

Iron-catalyzed trifluoromethylation of vinyl cyclopropanes: facile synthesis of CF₃-containing dihydronaphthalene derivatives

By: Li, Zong-Rui; et al

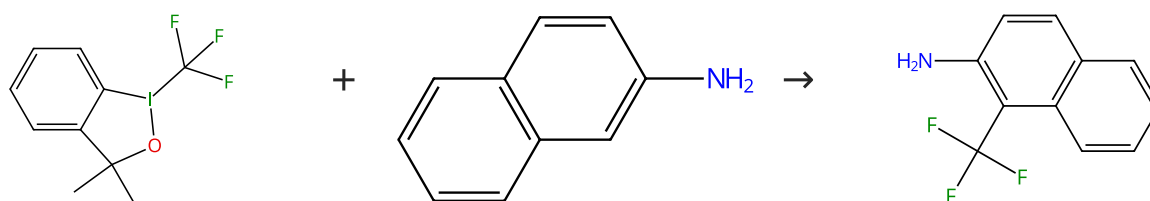
Organic Chemistry Frontiers (2016), 3(8), 934-938.

- 1.1 **Catalysts:** Ferrous chloride
Solvents: Dichloromethane; 10 - 30 min, 40 °C
- 1.2 **Reagents:** Potassium carbonate
Solvents: Water

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (67)

Suppliers (57)

Suppliers (17)

31-085-CAS-12413506

Steps: 1 Yield: 94%

Electrophilic trifluoromethylation of arenes and N-heteroarenes using hypervalent iodine reagents

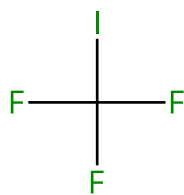
By: Wiehn, Matthias S.; et al

Journal of Fluorine Chemistry (2010), 131(9), 951-957.

- 1.1 **Reagents:** Chlorotris(trimethylsilyl)silane
Solvents: Acetonitrile; 24 h, 80 °C

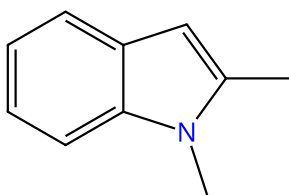
Scheme 35 (1 Reaction)

Steps: 1 Yield: 94%



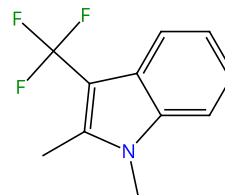
Suppliers (40)

+



Suppliers (64)

→



Suppliers (4)

31-085-CAS-1818500

Steps: 1 Yield: 94%

Trifluoromethylation of heterocycles via visible light photoredox catalysis

1.1 Reagents: *N,N,N,N*-Tetramethylethylenediamine
 Catalysts: Tris(2,2'-bipyridyl)ruthenium(II) chloride
 Solvents: Acetonitrile; 10 - 24 h, rt

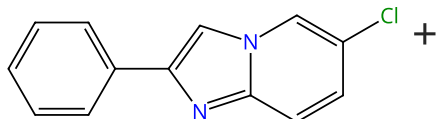
By: Iqbal, Naeem; et al

Tetrahedron Letters (2012), 53(15), 2005-2008.

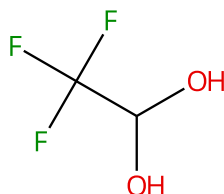
Experimental Protocols

Scheme 36 (1 Reaction)

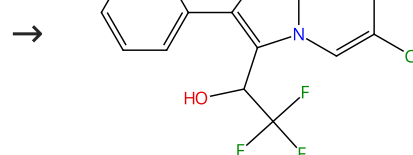
Steps: 1 Yield: 93%



Suppliers (57)



Suppliers (59)



31-614-CAS-39521388

Steps: 1 Yield: 93%

Metal-Free Synthesis of Trifluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines via Dehydrative Coupling of Imidazo[1,2-a]pyridines with Trifluoroacetaldehyde

1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

1.2 Solvents: Water; rt

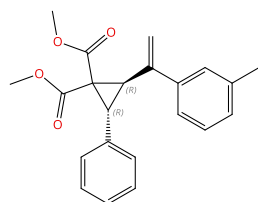
By: Liu, Zhaowen; et al

Synthesis (2024), 56(11), 1756-1764.

Experimental Protocols

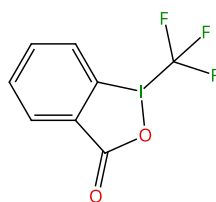
Scheme 37 (1 Reaction)

Steps: 1 Yield: 93%



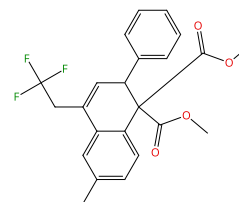
Relative stereochemistry shown

+



Suppliers (65)

→



31-085-CAS-17104258

Steps: 1 Yield: 93%

Iron-catalyzed trifluoromethylation of vinyl cyclopropanes: facile synthesis of CF₃-containing dihydronaphthalene derivatives

1.1 Catalysts: Ferrous chloride
 Solvents: Dichloromethane; 10 - 30 min, 40 °C

1.2 Reagents: Potassium carbonate
 Solvents: Water

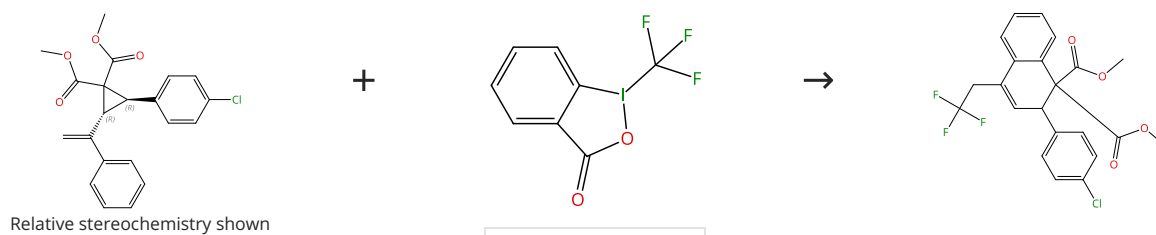
By: Li, Zong-Rui; et al

Organic Chemistry Frontiers (2016), 3(8), 934-938.

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (65)

31-085-CAS-17104261

Steps: 1 Yield: 93%

Iron-catalyzed trifluoromethylation of vinyl cyclopropanes: facile synthesis of CF₃-containing dihydronaphthalene derivatives

By: Li, Zong-Rui; et al

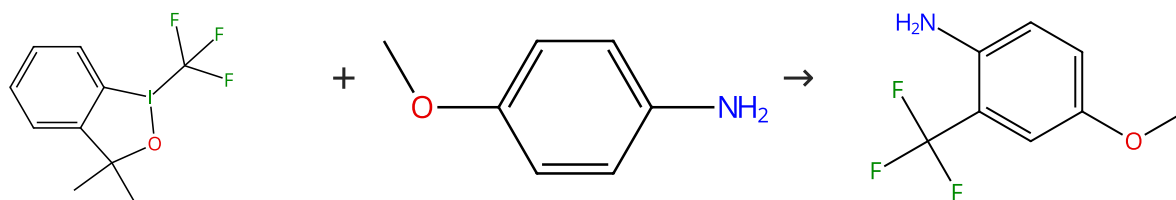
Organic Chemistry Frontiers (2016), 3(8), 934-938.

- 1.1 **Catalysts:** Ferrous chloride
Solvents: Dichloromethane; 10 - 30 min, 40 °C
- 1.2 **Reagents:** Potassium carbonate
Solvents: Water

Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (67)

Suppliers (83)

Suppliers (65)

31-085-CAS-22065862

Steps: 1 Yield: 93%

Transition metal-free direct C-H trifluoromethylation of (hetero)arenes with Togni's reagent

By: Chen, Xiaoyu; et al

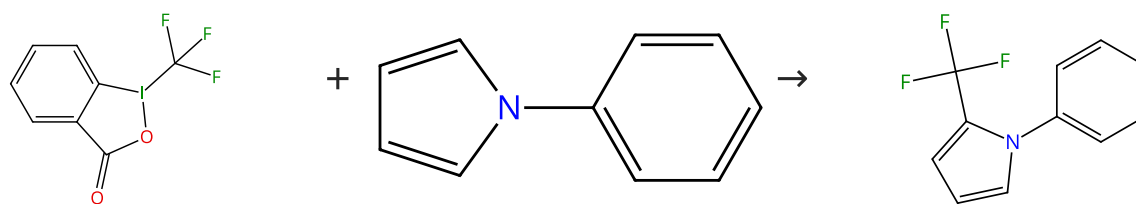
Tetrahedron Letters (2020), 61(9), 151538.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Acetonitrile; 6 h, 75 °C

Experimental Protocols

Scheme 40 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (65)

Suppliers (71)

Supplier (1)

31-614-CAS-33648195

Steps: 1 Yield: 93%

Direct C-H Trifluoromethylation of (Hetero)Arenes in Water Enabled by Organic Photoredox-Active Amphiphilic Nanoparticles

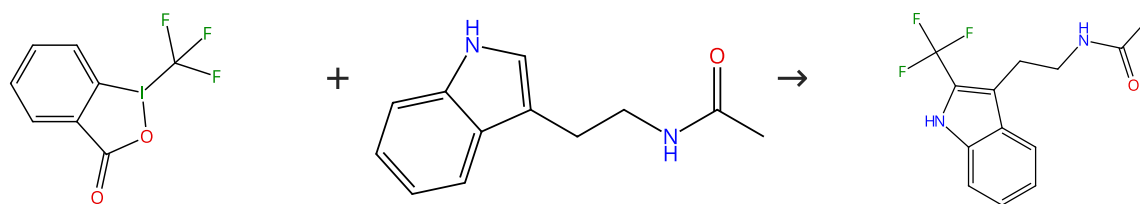
By: Eisenreich, Fabian; et al

Chemistry - A European Journal (2022), 28(52), e202201322.

- 1.1 **Catalysts:** Dodecylamine (reaction products with end-cap modified poly(pentafluorophenyl acrylat...), Jeffamine M 1000 (reaction products with end-cap modified poly(pentafluorophenyl acrylat...), 4-(10H-Phenothiazin-10-yl)benzeneethanamine (reaction products with end-cap modified poly(pentafluorophenyl acrylat...))
Solvents: Water; 3 h, 20 °C

Scheme 41 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (65)

Suppliers (70)

Suppliers (2)

31-085-CAS-11746606

Steps: 1 Yield: 93%

Direct C2-trifluoromethylation of indole derivatives catalyzed by copper acetate

By: Shimizu, Ryo; et al

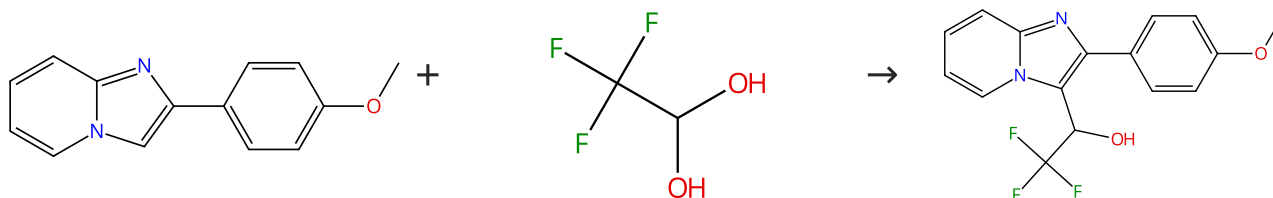
Tetrahedron Letters (2010), 51(45), 5947-5949.

 1.1 Catalysts: Cuprous acetate
 Solvents: Methanol; 24 h, rt

Experimental Protocols

Scheme 42 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (44)

Suppliers (59)

31-614-CAS-39521359

Steps: 1 Yield: 93%

Metal-Free Synthesis of Trifluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines via Dehydrative Coupling of Imidazo[1,2-a]pyridines with Trifluoroacetaldehyde

By: Liu, Zhaowen; et al

Synthesis (2024), 56(11), 1756-1764.

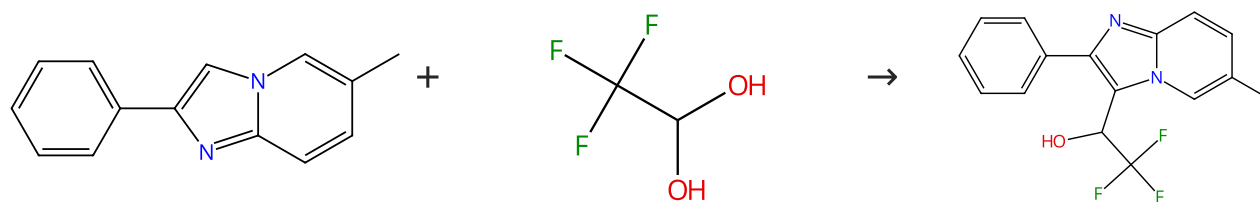
1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

1.2 Solvents: Water; rt

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (54)

Suppliers (59)

31-614-CAS-39521382

Steps: 1 Yield: 93%

Metal-Free Synthesis of Trifluoromethyl Carbinol-Containing Imidazo[1,2-a]pyridines via Dehydrative Coupling of Imidazo[1,2-a]pyridines with Trifluoroacetaldehyde

By: Liu, Zhaowen; et al

Synthesis (2024), 56(11), 1756-1764.

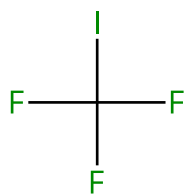
1.1 Solvents: 1,1,1,3,3,3-Hexafluoro-2-propanol; 12 h, rt

1.2 Solvents: Water; rt

Experimental Protocols

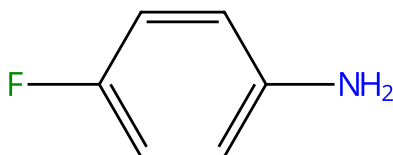
Scheme 44 (1 Reaction)

Steps: 1 Yield: 93%



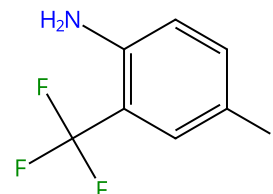
Suppliers (40)

+



Suppliers (70)

→



Suppliers (73)

31-085-CAS-18338693

Steps: 1 Yield: 93%

Visible-light-mediated direct perfluoroalkylation and trifluoromethylation of free anilines

1.1 **Reagents:** Potassium carbonate
Catalysts: *fac*-Tris(2-(2-pyridinyl)phenyl)iridium
Solvents: 1,2-Dichloroethane; 24 h, rt

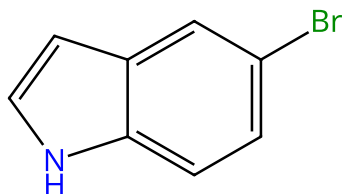
By: He, Chun-Yang; et al

Tetrahedron Letters (2017), 58(41), 3939-3941.

Experimental Protocols

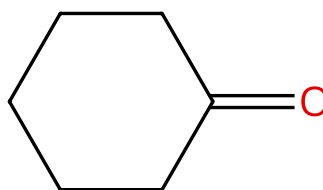
Scheme 45 (1 Reaction)

Steps: 1 Yield: 93%



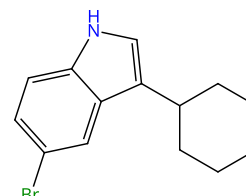
Suppliers (108)

+



Suppliers (102)

→



Suppliers (4)

31-085-CAS-5908388

Steps: 1 Yield: 93%

Palladium-Catalyzed Oxidative Trifluoromethylation of Indoles at Room Temperature

1.1 **Reagents:** Trichloroacetic acid, Triethylsilane
Solvents: Toluene; 2 h, 70 °C; 70 °C → rt

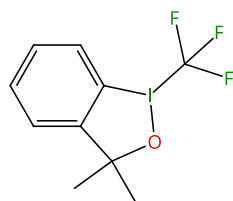
By: Mu, Xin; et al

Chemistry - A European Journal (2011), 17(22), 6039-6042.

Experimental Protocols

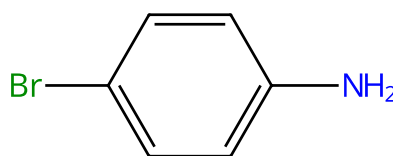
Scheme 46 (1 Reaction)

Steps: 1 Yield: 92%



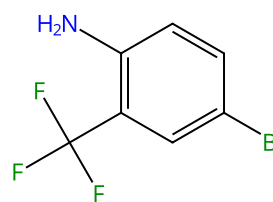
Suppliers (67)

+



Suppliers (83)

→



Suppliers (74)

31-085-CAS-22065855

Steps: 1 Yield: 92%

Transition metal-free direct C-H trifluoromethylation of (hetero)arenes with Togni's reagent

1.1 **Reagents:** Potassium carbonate
Solvents: Acetonitrile; 6 h, 75 °C

By: Chen, Xiaoyu; et al

Tetrahedron Letters (2020), 61(9), 151538.

Experimental Protocols